

Task 1: Ethernet vs Fiber Optic

Difference	Ethernet Cable	Fiber Optic Cable
Transmission	Uses electrical signals to transmit data.	Uses light signals to transmit data.
Speed & Distance	Suitable for high-speed connections over shorter distances.	Supports very high-speed connections over long distances.
Cost	Generally cheaper and easier to install.	Generally more expensive and requires specialized equipment.

- **Ethernet:** Used to connect PCs, printers, and switches in homes, offices, and gaming cafés.
- **Fiber Optic:** Used for high-speed backbone connections in data centers and college campuses.

Task 2: Straight-Through Ethernet Cable (T568B)

Pin	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Task 3: Steps to Attach an RJ45 Connector to a CAT6 Cable

- Cut the CAT6 cable to the required length.
- Remove a small part of the outer jacket from the cable.
- Separate and straighten the 8 internal wires.
- Arrange the wires in **T568B order**.
- Cut the wire ends evenly so they are the same length.
- Insert all 8 wires into the RJ45 connector.
- Make sure each wire reaches the correct pin inside the connector.
- Place the RJ45 connector into the **RJ45 crimper**.
- Squeeze the crimper firmly to attach the connector.
- Check the connector to make sure the wires are properly positioned.

Task 4: Choosing the Right Cable and Tool for a Gaming Cafe

For a gaming café with 10 PCs, I would choose **Ethernet cables** because they are fast, reliable, and affordable for connecting PCs over short distances. Ethernet cables are also easier to install and maintain. I would choose an **RJ45 crimper** because it can be used to attach RJ45 connectors to Ethernet cables. This setup is suitable for a gaming café because it provides good performance while keeping the installation cost low.